Example 10.17

If $X \sim U(200, 250)$ find its p.d.f and $P(X > 230)$

Solution :

(i) For $X \sim U(a, b)$

$$f(x) = \begin{cases} \frac{1}{b-a} & a < x < b \\ 0 & otherwise \end{cases}$$

Taking $a = 200$ and $b = 250$ the required p.d.f. is

$$f(x) = \begin{cases} \frac{1}{250-200} & 200 < x < 250 \\ 0 & otherwise \end{cases}$$

$$= \begin{cases} \frac{1}{50} & 200 < x < 250 \\ 0 & otherwise \end{cases}$$

(ii) $$P(x > 230) = \int_{230}^{\infty} f(x)dx = \int_{230}^{250} \frac{1}{50} dx$$

$$= \frac{1}{50}[x]_{230}^{250} = \frac{250-230}{50} = \frac{20}{50} = 0.4$$

Example 10.18

If X is a Uniform variate with the parameter 50 and100, find the mean, median and standard deviation.

Solution:

$X \sim U(50,100)$

Here $a=50$ $b=100$

$$\text{mean} = \text{median} = \frac{a+b}{2} = \frac{150}{2} = 75$$

$$\text{S.D} = \sqrt{\frac{(b-a)^2}{12}} = \frac{b-a}{\sqrt{12}} = \frac{100-50}{\sqrt{12}}$$

$$= \frac{50}{\sqrt{12}} = \frac{50}{3.464}$$

$$= 14.434$$

Example 10.19

If X is a random variable having a uniform distribution $U(a,b)$ such that $P(20<X<40)=0.2$ and mean = 150, find a and b.